Building $\Gamma^{(n)}$ from its symmetries instead of all 4^n elements

For a cluster of n sites the n-spin RDM is built element by element from the state $|\Psi\rangle$,

$$\Gamma^{(n)}_{\sigma',\sigma} = \Big\langle \Psi \Big| \prod_{k=1}^{n} d^\dagger_{k,\sigma'_k}\, d_{k,\sigma_k} \Big| \Psi \Big\rangle \equiv \langle \Psi | \hat{O}_{\sigma',\sigma} | \Psi \rangle, \qquad \rho_s = \frac{\Gamma^{(n)}}{\operatorname{Tr}\Gamma^{(n)}}, \tag{1}$$

where $\sigma = (\sigma_1, \ldots, \sigma_n)$ and σ' run over the 2^n spin configurations of the cluster. Computing every element costs 4^n expectation values per cluster. Three exact properties of $\Gamma^{(n)}$ reduce this number; none of them changes ρ_s.

1. S_z blocks. Let $d(\sigma)$ be the number of down spins in σ and $m(\sigma) = \frac{1}{2}\big(n - 2d(\sigma)\big)$ its S_z. The pair $d^\dagger_{k,\sigma'_k} d_{k,\sigma_k}$ removes a spin σ_k and adds a spin σ'_k on site k, so the product in Eq. (1) changes the total S_z by

$$[\hat{S}_z, \hat{O}_{\sigma',\sigma}] = \big(m(\sigma') - m(\sigma)\big)\, \hat{O}_{\sigma',\sigma}. \tag{2}$$

Applied to a state with definite $\hat{S}_z$ it gives a state with a different $\hat{S}_z$ unless $m(\sigma') = m(\sigma)$, and the overlap with $|\Psi\rangle$ vanishes:

$$\Gamma^{(n)}_{\sigma',\sigma} = 0 \qquad \text{if } d(\sigma') \neq d(\sigma). \tag{3}$$

Ordered by d, $\Gamma^{(n)} = \bigoplus_{d=0}^{n} \Gamma^{(n)}_{[d]}$ is block diagonal, with blocks of size $\binom{n}{d}$. A thermal state $\Gamma^{(n)}(T) = \sum_m p_m(T)\, \Gamma^{(n)}[\psi_m]$ commutes with $\hat{S}_z$ as well. Elements to compute: $\sum_d \binom{n}{d}^2 = \binom{2n}{n} \approx 4^n/\sqrt{\pi n}$.

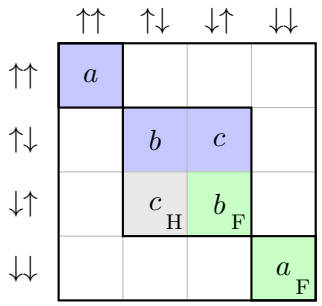
2. Hermiticity. $\hat{O}^\dagger_{\sigma',\sigma} = \hat{O}_{\sigma,\sigma'}$, hence $\Gamma^{(n)}_{\sigma,\sigma'} = \big(\Gamma^{(n)}_{\sigma',\sigma}\big)^*$. The Hubbard Hamiltonian is real, so $|\Psi\rangle$ and $\Gamma^{(n)}$ are real and $\Gamma^{(n)}_{\sigma,\sigma'} = \Gamma^{(n)}_{\sigma',\sigma}$. Only the diagonal and the upper triangle of each block are computed; the lower triangle is a copy.

3. Spin flip. Let $\hat{F}$ exchange the up and down orbital on every site, $\hat{F} d^\dagger_{k,\sigma} \hat{F}^\dagger = d^\dagger_{k,\bar{\sigma}}$, and let $\bar{\sigma}$ be σ with every spin flipped. Then $\hat{F}\hat{O}_{\sigma',\sigma}\hat{F}^\dagger = \hat{O}_{\bar{\sigma}',\bar{\sigma}}$. The Hamiltonian treats $\uparrow$ and $\downarrow$ alike, $[\hat{H}, \hat{F}] = 0$, and $\hat{F}$ maps the $S_z = 0$ sector onto itself, so a non-degenerate $S_z = 0$ eigenstate satisfies $\hat{F}|\Psi\rangle = \pm|\Psi\rangle$. Therefore

$$\Gamma^{(n)}_{\bar{\sigma}',\bar{\sigma}} = \langle \Psi | \hat{F}\hat{O}_{\sigma',\sigma}\hat{F}^\dagger | \Psi \rangle = (\pm 1)^2\, \Gamma^{(n)}_{\sigma',\sigma} = \Gamma^{(n)}_{\sigma',\sigma}. \tag{4}$$

The sign of $\hat{F}|\Psi\rangle$ cancels, so the relation holds for singlets and for $S_z = 0$ triplet components alike. It maps block d onto block $n - d$: only the blocks with $d \leq n/2$ are computed, and the middle block ($d = n/2$, even n) only up to the flip. Eq. (4) also holds for a ground state averaged over its degenerate manifold and for thermal states, since $[\hat{H}, \hat{F}] = 0$.

For $n = 2$ the configurations, ordered by d, are $\uparrow\uparrow$; $\uparrow\downarrow, \downarrow\uparrow$; $\downarrow\downarrow$. Of the 16 elements, 10 vanish by Eq. (3) and the remaining 6 carry only three numbers:



$$\rho_s = \begin{pmatrix} 0.1013 & 0 & 0 & 0 \\ 0 & 0.3987 & -0.2974 & 0 \\ 0 & -0.2974 & 0.3987 & 0 \\ 0 & 0 & 0 & 0.1013 \end{pmatrix}$$

Blue: computed (a, b, c). Grey, H: copy by hermiticity, a mirror across the diagonal; it never relates two diagonal elements. Green, F: copy by the spin flip, Eq. (4), which maps $\uparrow\uparrow \to \downarrow\downarrow$ and $\uparrow\downarrow \to \downarrow\uparrow$ and therefore also relates diagonal elements, $\Gamma^{(n)}_{\downarrow\downarrow,\downarrow\downarrow} = \Gamma^{(n)}_{\uparrow\uparrow,\uparrow\uparrow}$ and $\Gamma^{(n)}_{\downarrow\uparrow,\downarrow\uparrow} = \Gamma^{(n)}_{\uparrow\downarrow,\uparrow\downarrow}$. Right: the stored ρ_s for $L = 6$, $U/t = 10$, ground state, in the same order; the equal entries are exactly the pairs above.

Together the three properties leave about one twentieth of the elements:

n	all, 4^n	S_z blocks	+ hermiticity	+ spin flip	saving	$L = 10$, one cluster†
5	1 024	252	142	71	14×	2.9 min → 12 s
6	4 096	924	494	252	16×	12 min → 43 s
7	16 384	3 432	1 780	890	18×	46 min → 2.5 min
8	65 536	12 870	6 563	3 299	20×	3.1 h → 9 min
9	262 144	48 620	24 566	12 283	21×	12 h → 35 min

† cost of the present element-by-element construction at $L = 10$, $U/t = 10$ (sector dimension 63 504): 0.17 s per element.

For $n = L$ no expectation values are needed: ρ_s is the pure state $\hat{P}|\Psi\rangle$, with $\hat{P}$ the projector on singly occupied sites. On the ring a single cluster suffices, because the cluster average equals one cluster for a translation-invariant state (10^{-17} at $L = 6$).

In practice: group the configurations by $d(\sigma)$; compute $\Gamma^{(n)}_{\sigma',\sigma}$ with $\sigma' \le \sigma$ in the blocks $d < n/2$, and in the block $d = n/2$ one pair of each $\{(\sigma', \sigma), (\bar{\sigma}', \bar{\sigma})\}$; fill the lower triangles by hermiticity and the blocks $d > n/2$ by Eq. (4); set all elements between different blocks to zero; normalise and pass the full $2^n \times 2^n$ matrix ρ_s to the GCG.

The three properties were checked on the stored RDM bank ($L = 6$, $S_z = 0$ sector, $U/t = 4$ and 10, ground state and first excited state, $n = 2, \ldots, 5$). Every element with $d(\sigma') \neq d(\sigma)$ is exactly 0 (also for $n = 6$); $\max|\Gamma^{(n)} - (\Gamma^{(n)})^{\mathsf{T}}| \le 4 \times 10^{-17}$ and $\mathrm{Im}\,\Gamma^{(n)} = 0$; $\max|\Gamma^{(n)}_{\bar{\sigma}',\bar{\sigma}} - \Gamma^{(n)}_{\sigma',\sigma}| \le 8 \times 10^{-15}$, while $\max|\Gamma^{(n)}_{\bar{\sigma}',\bar{\sigma}} + \Gamma^{(n)}_{\sigma',\sigma}| \ge 0.24$, so the sign in Eq. (4) is +.

The convex roof $\mathcal{C}_{\mathrm{GME}}(\rho_s)$ cannot be split into blocks: the optimal decomposition $\rho_s = \sum_i p_i |\Phi_i\rangle\langle\Phi_i|$ may mix different S_z. The symmetries only reduce the construction of $\Gamma^{(n)}$ (and the eigendecomposition of ρ_s); the GCG still receives the full $2^n \times 2^n$ matrix.